

(KTXFI2EBNF) Physics I. Exam 1 Solutions
May 26, 2026

I. Mechanics

1. What does Newton's first law state? What does Newton's second law describe? What is Newton's third law? (5 points)
Lecture on February 16, 2026, Slide 17.
2. What are conservative forces? What are dissipative forces? (5 points)
Lecture on February 16, 2026, Slide 6.
3. What is momentum? What is conservation of momentum? (5 points)
Lecture on February 16, 2026, Slides 15, 18.
4. What is escape velocity? (5 points)
Lecture on February 16, 2026, Slide 31.

II. Thermodynamics

5. What does the first law of thermodynamics state? What is the second law of thermodynamics? (5 points)
Lecture on March 16, 2026, Slide 11.
6. What is thermal expansion? What is linear expansion? What is volumetric expansion? (5 points)
Lecture on March 09, 2026, Slides 5, 7.
7. What is the ideal gas model? What is the ideal gas equation conceptually? (5 points)
Lecture on March 09, 2026, Slides 20, 21.
8. What is the Carnot cycle? (5 points)
Lecture on March 23, 2026, Slide 9.

III. Electrostatics, Waves, and Optics

9. What is Coulomb's law? (5 points)
Lecture on April 13, 2026, Slide 13.
10. What is the Doppler effect? (5 points)
Lecture on March 02, 2026, Slide 32.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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